

Discussion of Professor Minge Xie's “Repro Samples Method...”¹

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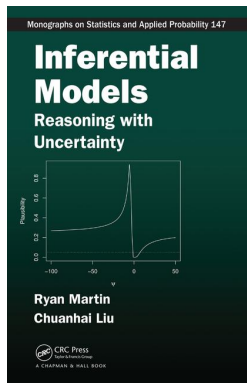
June 28th, 2023

¹Prof. Xie's paper: <https://arxiv.org/abs/2206.06421>

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- *Repro samples method*: Strategy for constructing confidence sets with provable coverage probability bounds
- Workhorse: “probability carrying” sets, B_α or $B_\alpha(\theta)$
- Coverage guarantees implied by pushing these “probability carrying” sets through data + model
- Nice Monte Carlo strategy for computing
- Impressive collection of applications — *structure learning*
- My focus: high-level connections...

- **Idea credit to BFF**: Fiducial inversion, IM, conformal prediction, ABC, etc.



- IMs offer reliable (imprecise-)probabilistic inference without priors
 - reliable $\longleftrightarrow$ frequentist
 - probabilistic $\longleftrightarrow$ Bayes
- Efron's "Holy Grail of Statistics"
- Imprecision is necessary to achieve the specific property we call *validity*
- Validity implies frequentist coverage
- Validity achieved³ by pushing a *random set* that's "stochastically probability carrying" through data + model

³Somewhat different approach in M. (JASA 2015, arXiv:1203.6665) with lots of new developments in M. arXiv:2211.14567

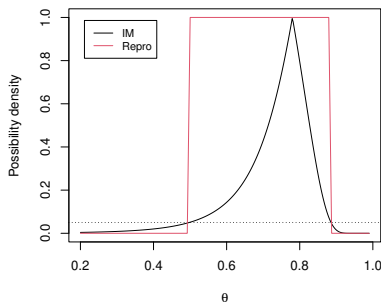
- My view: *repro is an advancement in IMs*
- “Beginner version” of repro is a special case of IMs
 - take random set for U to be degenerate at B_α
 - this simple case can go unexpectedly far
- “General version” is similar
 - take random set for $T(U, \theta)$ degenerate at $B_\alpha(\theta)$
 - we used similar ideas in various places⁴

The following corollary states that the validity theorem remains true even after a suitable—possibly θ -dependent—change of auxiliary variable. In other words, the validity property is independent of the choice of auxiliary variable parameterization. This reparameterization comes in handy in examples, including those in Section 5.

- but we didn’t develop the idea systematically
- Repro efforts show how broadly powerful this idea is!

⁴M. and Liu (JASA 2013, arXiv:1206.4091)

- Example:
 - standard bivariate normal, unknown correlation θ
 - *localized IM* uses θ -dependent random sets⁵
- A repro solution: ours but w/ degenerate random set
- Both use a version of DS-calculus, return imprecise prob's
- IM & repro “posterior possibility densities” ($\alpha = 0.05$)



⁵M. and Liu (*JRSS-B* 2015, [arXiv:1211.1530](#))

Remark: embrace imprecision!

An imprecise-probabilistic characterization of
frequentist statistical inference*

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From Theorem 6 in M. [arXiv:2112.10904](https://arxiv.org/abs/2112.10904)

Let $\{C_\alpha : \alpha \in [0, 1]\}$ be any family of nominal confidence sets. Under very mild conditions, there exists a valid *fused IM*^a such that the derived confidence sets C_α^{IM} satisfy

$$C_\alpha^{\text{IM}}(y) \subseteq C_\alpha(y), \quad \text{all } y \text{ and all } \alpha.$$

^a*Fused IM* = IM w/ parameter-dependent random set $\leftrightarrow$ repro

A sort of complete-class theorem for repro?

- These advancements/connections open new doors
- There's been recent IM progress in ML-style problems⁶ but repro ought to help guide IM structure learning
- Some specific questions/suggestions:
 - 1 In, say, location problems, it makes sense to condition on ancillary statistics — conditioning in the repro framework?
 - 2 Repro solutions for some classical nuisance parameter problems, e.g., Behrens–Fisher?⁷
 - 3 Inference on max-of-means (re: DTR) — repro solution?
 - 4 Mixture models — isn't a confidence lower bound for the # of components all one wants/all the data can provide?
 - 5 High-dim reg — “no minimal signal size requirement”?

⁶e.g., Cella & M. (*IJAR* 2022ab; arXiv:2112.10234, arXiv:2112.10232)

⁷M. and Liu (*JASA* 2015, arXiv:1306.3092)

Thanks!

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ST 790 (001) Fall 2022 Advanced Special Topics

Imprecise-Probabilistic Foundations of Statistics & Data Science

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